



An efficient master head selection for multi-EEG to multi-fog IoT network using 6G-driven FaaS

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ABSTRACT

An Electroencephalogram (EEG) signal plays a vital role in a healthcare communication system for recording the electrical activities of the human brain from the scalp. In recent times, the conventional IoT-based healthcare system uses the cloud computing paradigm to manage time-critical healthcare data. Moreover, switching to the fog computing, the fog-assisted EEG systems are for single EEG applications. However, the use of a fog computing paradigm for a single EEG system is not an efficient solution in terms of resource management and time consumption. Therefore, we introduce a Fog-enabled EEG architecture where multiple fog devices collaboratively process the data in a single integrated IoT platform. As the proposed architecture is new, we focus on developing the mathematical model of the architecture and discuss the crucial aspects. Additionally, we devise a dynamic optimal fog head selection within the network using a weighted multi-criteria decision-making approach. From the simulation, we observe that the average propagation delay is reduced by approximately 95% using 6G-enabled fog computing as compared to the cloud. Further, our method reduces the total delay by 83.87% compared to the existing baseline KCHE technique, showing the effectiveness of this work.

1. Introduction

Nowadays, advancement in Internet of Things (IoT) technologies is laying the foundation for the development of 6G communication systems. IoT with 6G is crucial for modern human life and has garnered significant attention from both academia and industry due to its enhanced potential customer services. The superior features of 6G, such as ultra low latency, higher data rate, and extremely high throughput are expected to offer unprecedented service quality and elevate user experiences in current IoT systems [1,2]. Standardization and deployment of 6G are anticipated by 2030, shifting research focus towards these next-generation communication systems [3,4]. The transition to 6G will bring revolutionary advancements in smart devices, technologies, artificial intelligence, autonomous vehicles, wireless Brain-Computer Interfaces (wBCI), and so on [5,6]. During this paradigm shift, IoT has a critical role in enabling emerging healthcare applications by connecting healthcare devices in the physical environment to wireless communication and computing systems in the network environment. These IoT-enabled healthcare applications demand superior performance in terms of data rate, latency, coverage, and resource optimization [7]. Consequently, 6G ensures seamless connectivity for millions of healthcare and computing devices and applications, supporting massive inter

connectivity in IoT with diverse service requirements. In this aspect, brain-computer interface has become a cornerstone of the healthcare sector in the past decade, revolutionizing the way human brains and machines communicate. The signals generated from the human brain, known as EEG signals, are data-intensive and require real-time analysis [8]. With the evolution of healthcare IoT, the communication between EEG devices and computation-intensive devices has advanced significantly, particularly in time-critical scenarios. The ever-increasing demand for high data rates in EEG Healthcare Systems (EHS) has been the main driving force for this evolution. There have been technical advancements in EEG devices in the past few years, such as improved electrode technology for improving signal quality, bluetooth and Wi-Fi connectivity for real-time data transmission, advanced Filtering Techniques for noise reduction and signal enhancement, and integration with wearable to provide comprehensive health monitoring. They generate massive volumes of data, creating significant traffic for processing devices and transforming today's healthcare IoT into a future-ready IoT. It facilitates seamless communication between heterogeneous devices without human intervention, potentially revolutionizing healthcare industries and offering significant societal benefits. These unprecedented levels of capacity accelerate the applications of EEG signal processing

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deployment of 6G-enabled IoT networks across healthcare data sensing, healthcare device connectivity, wireless communication with processing devices, and 6G network management. Unlike previous mobile network generations, 6G offers larger communication channels and can handle the explosive traffic growth that existing mobile networks struggle with due to capacity limitations in radio access, backhaul, and centralized cloud networks [9].

1.1. Motivation

In the existing EHS, the large volume of data generated by EEG devices is primarily reliant on cloud services or on peer-to-peer communication, where each standalone EEG device interacts with a dedicated fog device [10,11]. EEG devices connected to a cloud server are sometimes inefficient because of four major challenges — significant requirements of network bandwidth, additional computational latency, prolonged end-to-end delays, and high costs of centralized cloud maintenance [12,13]. These challenges are critical in IoT-enabled EEG applications, which often demand real-time operations with decentralized services. Therefore, fog computing addresses these issues effectively due to its inherent features such as distributed nature, significantly lower latency and bandwidth compared to centralized cloud computing, and providing real-time processing capabilities essential for healthcare applications [14]. However, existing EHS that rely on dedicated fog computing services primarily use single-device single fog or single-device multi-fog architectures. These configurations often lead to suboptimal resource utilization and increased costs such as over utilization caused by the large volume of data, under utilization during non-critical or limited-time operations, and complete idleness when the associated EEG device is not in use. Again, for multi-EEG multi-fog platform, randomly assigning tasks to any fog node leads to the following problems — unequal task distribution, redundant computations of EEG tasks, lack of synchronization and time-aligned aggregation of EEG responses, and addressing heterogeneity. Additionally, with a single master head, the inter-fog communication complexity is reduced [15], which is critical for real-time EEG analytics. Furthermore, 5G or earlier generations struggle to support most of the advanced technologies and sophisticated healthcare applications and often reaches its limits in terms of scalability [16]. To overcome these challenges, we explore the potential of a 6G-enabled IoT network aiming to satisfy the requirements for a multi-EEG multi-fog paradigm, ensuring speed and immediacy in the response to the EHS.

1.2. Contribution

This paper proposes a 6G-enabled fog computing distributed architecture to handle healthcare data, specifically the Electroencephalogram signals in an Internet of Things network. The major contribution of this work is fourfold:

1. We design a novel $\mathbb{E}2\mathbb{F}$ architecture that enables multichannel, multiple EEG devices to connect to a 6G-coupled fog layer within a healthcare IoT network. By leveraging Fog-as-a-Service (FaaS), our architecture supports heterogeneous fog devices, facilitating collaborative and distributed processing of EEG data. Additionally, we provide an in-depth discussion of each component of our architecture.
2. Further, in this work, the challenge posed by the diverse specifications and computational capabilities of heterogeneous fog devices is addressed. To tackle this, we design a system model for the $\mathbb{E}2\mathbb{F}$ architecture, incorporating a comprehensive set of parameters from all relevant components, such as EEG-related, fog-related, and network-related parameters.

3. This work introduces an optimal selection mechanism for designating a master head that is responsible for effective task distribution and workload management among fog devices. The mechanism incorporates 6G-aware network parameters to enhance efficiency. Moreover, we implement the method of criteria importance through an inter-criteria correlation approach to ensure the unbiasedness of this approach.
4. For our $\mathbb{E}2\mathbb{F}$ architecture, we further analyze the performance of this architecture across a hundred iterations in the experimental simulation by varying the key influencing parameters such as number of devices, number of EEG tasks, fog clock frequency, fog energy, and network bandwidth.

The rest of the paper is organized as follows. Section 2 describes the existing work using fog computing and IoT for EEG healthcare systems. Section 3 outlines different parameters and the system modeling of $\mathbb{E}2\mathbb{F}$ architecture. Moreover, this section defines the problem formulation and the objective function. Section 4 presents the proposed master head selection technique. Additionally, the proposed technique is demonstrated with an illustrated example. Section 5 highlights the performance analysis and comparative analysis of this work. Finally, Section 6 concludes the paper.

2. Related work

The existing works are categorized into three parts. The first part describes the revolutionary development in the EEG-based Brain Computer Interaction (BCI) applications for the need of 6G-aware fog environment in HIoT. The second part describes the scope of fog computing in EEG-based healthcare systems. In third part, we highlight a comparative analysis of EHS architectures and outline the research questions.

In [17], Jafri et al. have implemented machine learning algorithms for smart home and medical systems utilizing the EEG-based BCI, demonstrating how brain signals can be translated into commands for external digital devices. Such applications, which rely on rapid and seamless brain-device interactions, require wireless communication technologies to overcome the limitations of bulky and complex wired interfaces. In [18], Fang et al. have developed emotion recognition system using deep learning technique multi-channel EEG signals, ranging from delta (1–4 Hz) to gamma (30–50 Hz), processed in an edge AI environment. The real-time EEG data classification took milliseconds during the training, whereas, many medical and smart monitoring applications demand ultra-low latency for interpretation of commands and responses, potentially in micro or nanoseconds. Furthermore, the inherently low amplitude of EEG signals necessitates highly reliable, accurate, and robust wireless transmission across IoT networks. Emerging 6G technologies can meet these requirements by enabling high-throughput, ultra-reliable, and low-latency remote monitoring for neurological and sensor-based healthcare applications [19–21]. On the other side, the massive, multi-channel EEG data streams are highly time-sensitive and resource-intensive. To handle this, IoT architectures increasingly rely on fog-assisted environments that distribute computation and minimize response times while maintaining Quality-of-Service (QoS). In this direction, Long et al. [22] and Fei Xu et al. [23] have proposed collaborative task offloading strategies combining edge and cloud computing, which optimize both energy efficiency and delay, with 6G-enabled edge computing showing notably faster convergence. Similarly, Rahmani et al. [24] have demonstrated how clustered fog device structures, interconnected through wireless links to IoT devices, improve task processing—but at the cost of increased communication overhead and system complexity. The integration of fog computing in EEG-based healthcare systems has yielded significant advancements, yet these findings reveal that effective task allocation within the fog layer remains an NP-hard problem, primarily due to heterogeneity in resources and the complexity of distributed coordination [25]. Analyzing previous work highlights, the authors are motivated to focus on

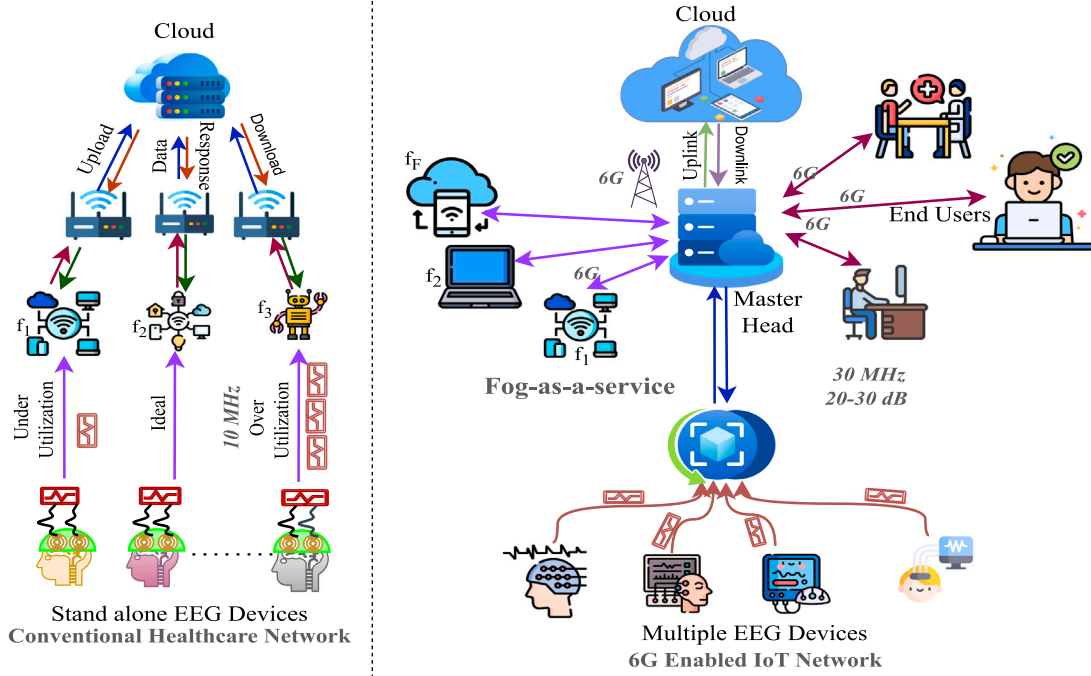


Fig. 1. Proposed E2F architecture for multiple EEG devices and 6G-aware multiple fog devices in a healthcare IoT network.

addressing two main research questions - (i) *How to adopt standardization, interoperability, and diversity of devices of fog computing solutions for stand-alone multiple multichannel EEG devices?* (ii) *How to optimum substantial computational resources for real-time processing of EEG signals through the selection of the best fog device involving all EEG, fog and network parameters?* To address these issues, we come up with E2F technique that selects an optimal distinguished master head considering several key advantages such as efficient handling of heterogeneous fog resources, provision of centralized cloud-like services in closer proximity to EEG-based healthcare devices, minimizing both delay and energy consumption by adapting the advanced features of 6G networks, and enabling distributive and collaborative processing through the master head. Table 1 gives a brief comparative description of EHS with different IoT configurations.

We specifically address the research question - *How can 6G capabilities be leveraged to enhance multi-fog collaboration of the master-head selection process in EEG-driven IoT systems?* Although the proposed approach supports any advanced fog-enabled IoT network, 6G provides ideal conditions for multi-fog collaboration due to its high bandwidth capability, low latency response, and reliable device connectivity. EEG healthcare systems often involve dense deployments of wearable sensors. 6G's support for massive machine-type communications (mMTC) ensures stable connectivity for multiple EEGs with multiple fog devices simultaneously [26]. This strengthens the scalability of the proposed architecture, particularly when multiple EEGs generate concurrent tasks. Furthermore, 6G supports fog-to-fog and fog-to-cloud offloading through hierarchical computing [27]. The proposed master head selection in this work benefits significantly by reducing overall service delay in sub-millisecond. Moreover, 6G offers distributed intelligence for fog collaboration and improves the precision of transmission delay estimation. Therefore, the architecture can be generalized to advanced fog-IoT wireless systems while leveraging 6G features for optimal performance.

3. System model and problem formulation

The architecture shown in Fig. 1 is a network of collections of multiple EEG devices and multiple fog devices. The EEG devices are

independent and the fog devices are heterogeneous in nature. Therefore, the configurations and capabilities of the devices vary from one to another. We focus on the mathematical modeling of E2F architecture considering major parameters of EEGs and fogs. The selection of the master node and communication between the base layer and the service layer are modeled as follows.

3.1. Network design

Let us consider 3 sets- $\mathcal{F}$, $\mathcal{E}$, and $\mathcal{T}$ to denote a set of fogs, EEGs, and tasks, respectively. The set of heterogeneous fog devices is represented as, $\mathcal{F} = \{f_i \mid f_i \text{ is a fog device, } i \in \{1, 2, 3, \dots, F\}\}$, where, $|\mathcal{F}| = F$, $\forall f \in \mathcal{F}$. The set of independent EEG devices is represented as, $\mathcal{E} = \{e_j \mid e_j \text{ is an EEG device, } j \in \{1, 2, 3, \dots, E\}\}$, where, $|\mathcal{E}| = E$, $\forall e \in \mathcal{E}$. The set of EEG Tasks generated by the EEG devices may or may not equal and is represented as, $\mathcal{T} = \{t_{jk} \mid t_{jk} \text{ is } k^{\text{th}} \text{ task of } j^{\text{th}} \text{ EEG, } k \in \{1, 2, 3, \dots, T\}\}$, where, $|\mathcal{T}| = T$, $\forall t \in \mathcal{T}$. The different parameters that are considered for the mathematical modeling of this architecture are divided into three parts: fog, EEG, and network-related parameters.

- The fog devices that participate in this system architecture have different configurations because of the variations in parameters such as minimum clock frequency ($c^{\min}$), maximum clock frequency ($c^{\max}$), number of cores in the processor (ζ), Million Instructions per Second ($\mathcal{M}$), primary storage (S^{pri}), secondary storage (S^{sec}), and battery capacity in the form of energy (E).
- The EEG parameters taken into consideration are number of channels (N), number of tasks (T) and size of each task ($\mathcal{V}(t)$).
- Further, the network-based parameters are signal-to-noise ratio (SNR), network bandwidth (B), and data propagation speed in the medium (S), coordinates of EEG (x_e, y_e), and coordinates of fog (x_f, y_f).

The selection of the master head depends on the four factors- (i) Fog Service Delay ($\mathbb{D}$), (ii) Fog Residual Energy ($\mathbb{R}$), (iii) Fog Memory Efficiency ($\mathbb{M}$), (iv) Fog Performance Efficiency ($\mathbb{P}$).

Table 1

Comparison of various architecture based EHS.

Features	Architecture 1 (A1) Cloud Computing [28–30]	Architecture 2 (A2) Peer-to-Peer Dedicated Fog Computing [31,32]	Architecture 3 (A3) Proposed Multiple Fog Through Optimal Master Head [33,34]
Overall Delay	High, due to more propagation and queuing delay	Minimal but additional space and cost	Minimal
Bandwidth Usage	High, due to huge data transmission volume	Lower than A1 but higher than A3 (all fogs connect to cloud)	Minimal, as master head aggregates data before transmission
Energy Consumption	High, devices use energy for continuous long-range data transmission	Less, single point connection	Less, single point connection
Cost	High, due to heavy processing, storage, and transmission	Higher than A3, maintaining dedicated fog for each EEG	Cheaper than A1 and A2, distributed multi-fog processing
Scalability	Very high (virtually unlimited cloud resources)	Very low (dedicated fogs can overload)	High (multi-fog with master head management)
Complexity	Low (cloud handles everything)	Medium–high (each fog links to EEG + cloud)	Medium–high (depends on network size, requires monitoring)
Real-time Performance	Not preferred for critical EEG (due to delays)	Suitable for critical EEG but may suffer from overload or underutilization	Near real-time, master fog ensures fair distribution
Fault Tolerance	Low (if cloud link fails, service stops)	Medium (fog failure leads to data loss)	High (alternate master head + multi-fog task offloading)
Services	Centralized	Dedicated	Centralized with distributed support system
Best Use Case	Long-term storage, offline analysis, non-critical EEG apps	Localized EEG monitoring (small or medium deployment)	Large-scale EHS

Definition 1. The Fog Service Delay ($\mathbb{D}_{f_i}$) of i th fog device is defined as the sum of two delays - (i) total network delay ($\mathbb{D}_{f_i}^{\text{net}}$) and (ii) total computational delay ($\mathbb{D}_{f_i}^{\text{comp}}$). Total network delay is the collective time of $\mathbb{D}_{f_i}^{\text{trans}}$ and $\mathbb{D}_{f_i}^{\text{prop}}$ and total computational delay is the sum of $\mathbb{D}_{f_i}^{\text{proc}}$ and $\mathbb{D}_{f_i}^{\text{que}}$.

We evaluate the overall delay of a fog device in our architecture in two parts. The first part derives the network delay considering the EEG data transmission time and the data propagation time as given in Eq. (1). The second part derives the computational delay considering processing time and average waiting time in the queue as given in Eq. (2).

$$\mathbb{D}_{f_i}^{\text{net}} = \mathbb{D}_{f_i}^{\text{trans}} + \mathbb{D}_{f_i}^{\text{prop}} \quad (1)$$

$$\mathbb{D}_{f_i}^{\text{comp}} = \mathbb{D}_{f_i}^{\text{proc}} + \mathbb{D}_{f_i}^{\text{que}} \quad (2)$$

From Eqs. (1) and (2), the fog service delay is derived as,

$$\mathbb{D}_{f_i} = \mathbb{D}_{f_i}^{\text{net}} + \mathbb{D}_{f_i}^{\text{comp}} \quad (3)$$

where, $\mathbb{D}_{f_i}^{\text{trans}}$ refers to transmission delay, $\mathbb{D}_{f_i}^{\text{prop}}$ refers to propagation delay, $\mathbb{D}_{f_i}^{\text{proc}}$ refers to processing delay, and $\mathbb{D}_{f_i}^{\text{que}}$ refers to average queuing delay.

The data transmission time indicates the time taken for the total EEG data volume to upload in the transmission medium. It is defined as the ratio of sum of all the EEG tasks to the maximum uploading bit rate.

$$\mathbb{D}_{f_i}^{\text{trans}} = \frac{\sum_{j=1}^E \sum_{k=1}^{T_j} \mathcal{V}(t_{jk})}{\mathcal{R}} \quad (4)$$

In Eq. (4), T_j refers to the number of tasks of j th EEG, which is assumed to be variable from one EEG device to the other. For a band-limited channel B in Hz, let $\mathcal{R}$ be the channel capacity between transmitted signal D^T and received signal D^R . Assuming a Gaussian noise channel with noise random variable D^X , let the signal and noise power received in the wireless medium be $\mathcal{N}_{\text{db}}^{\text{wilsig}}$ and $\mathcal{N}_{\text{db}}^{\text{wlnos}}$ in db respectively. By Shannon's theorem, we define the mutual information $I(D^T : D^R)$ as the differential entropy of D^R and D^X as $I(D^T : D^R) = \xi(D^R) - \xi(D^X)$, where $D^R = D^T + D^X$ [35]. Therefore, using Nyquist sampling theorem, the maximum bit rate is determined by the channel capacity of the transmission medium considering network bandwidth and the signal-to-noise ratio, as given in Eq. (5).

$$\mathcal{R} = B \cdot \log_2 \left(1 + \frac{\mathcal{N}_{\text{db}}^{\text{wilsig}}}{\mathcal{N}_{\text{db}}^{\text{wlnos}}} \right) \quad (5)$$

Definition 2. Let $N(e_j)$ be the number of channels of j th EEG device, $F_s(e_j)$ be the sampling frequency (in samples per second), and $S_t(e_j)$ be the time stamp (in seconds). The size of k th task of j th EEG is calculated as the product of number of channels, sampling frequency and the time stamp. Therefore, $\mathcal{V}(t_{jk}) = N(e_j) \times F_s(e_j) \times S_t(e_j)$.

In $\mathbb{E2F}$ architecture, the EEG data generated in the base layer is transmitted to the service layer for processing. Therefore, the data travel time from EEG devices to the fog devices is a necessary parameter, referred to as data propagation time. This is calculated as the distance from a fog device to the sum of the average distance between EEG devices ($\mathcal{D}_{f_i,E}^{\text{avg}}$) and the rest of the fog devices ($\mathcal{D}_{f_i,\bar{F}}^{\text{avg}}$). Moreover, the propagation speed of the signal S in the wireless medium is equivalent to the speed of light.

$$\mathbb{D}_k^{\text{prop}} = \frac{\mathcal{D}_{f_i,E}^{\text{avg}} + \mathcal{D}_{f_i,\bar{F}}^{\text{avg}}}{S} \quad (6)$$

In Eq. (6), the distance between EEG devices and the fog devices is calculated using Euclidean formulation, given the coordinates of i th fog and j th EEG as (x_{f_i}, y_{f_i}) and (x_{e_j}, y_{e_j}) respectively.

$$\mathcal{D}_{f_i,E}^{\text{avg}} = \left(\frac{1}{E} \right) \sum_{j=1}^E \sqrt{(x_{f_i} - x_{e_j})^2 + (y_{f_i} - y_{e_j})^2} \quad (7a)$$

$$\mathcal{D}_{f_i,\bar{F}}^{\text{avg}} = \left(\frac{1}{F-1} \right) \sum_{i=1}^{F-1} \sqrt{(x_{f_i} - x_{\bar{f}_i})^2 + (y_{f_i} - y_{\bar{f}_i})^2} \quad (7b)$$

where, $f_i \neq \bar{f}_i, \forall \bar{f}_i \in \mathcal{F}'$ and $\mathcal{F}' \subset \mathcal{F}$. Further, $|\mathcal{F}'| = \bar{F}$.

The processing time of an EEG task implicitly depends on the clock frequency of a fog device. A higher clock frequency of the device means less processing time for the task. The average clock frequency varies from device to device. We calculate the processing time of total tasks for i th fog device as,

$$\mathbb{D}_{f_i}^{\text{proc}} = \frac{\sum_{j=1}^E \sum_{k=1}^{T_j} \mathcal{V}(t_{jk})}{c_{f_i}^{\text{avg}}} \quad (8)$$

where, $c_{f_i}^{\text{avg}}$ is the average clock frequency defined in Eq. (9)

$$c_{f_i}^{\text{avg}} = \frac{c_{f_i}^{\text{min}} + c_{f_i}^{\text{max}}}{2} \quad (9)$$

The EEG devices generate continuous signals, resulting in a huge volume of data. On the other hand, fog devices have limited storage

capabilities to handle the EEG data. While certain EEG tasks are in process with the fog devices, the rest of the tasks must wait in the device queue, considering unwanted waiting time, referred to as queuing delay. We assume the following conditions to calculate queuing delay:

1. A particular EEG task can be assigned to only one fog device at a time.
2. A fog device can only process one EEG task at a time.
3. The first ever EEG task comes for processing and has zero queuing delay considering all fog devices.

Definition 3. The average queuing delay for n th task assigned to i th fog device is the sum of processing times taken by 2nd task up to $(n-1)^{th}$ task, assuming the queuing delay for 1st task is zero.

Let $\mathcal{T}_{f_i}$ represent the set of tasks assigned to i th fog device from multiple EEG devices. where, $\mathcal{T}_{f_i} \subseteq \mathcal{T}$ and $|\mathcal{T}_{f_i}| = n$. The queuing delay for n th task is,

$$\mathbb{D}_{f_i}^{\text{que}}(t_n) = \sum_{k=2}^{n-1} \mathbb{D}_{f_i}^{\text{proc}}(t_k) \quad (10)$$

From Eq. (10), we have Eq. (11) as,

$$\begin{aligned} \sum_{k=1}^n \mathbb{D}_{f_i}^{\text{que}}(t_k) &= \mathbb{D}_{f_i}^{\text{proc}}(t_1) + (\mathbb{D}_{f_i}^{\text{proc}}(t_1) + \mathbb{D}_{f_i}^{\text{proc}}(t_2)) \\ &+ (\mathbb{D}_{f_i}^{\text{proc}}(t_1) + \mathbb{D}_{f_i}^{\text{proc}}(t_2) + \mathbb{D}_{f_i}^{\text{proc}}(t_3)) \\ &+ \dots + \left(\sum_{k=2}^{n-1} \mathbb{D}_{f_i}^{\text{proc}}(t_k) \right) \end{aligned} \quad (11)$$

The total average queuing delay at i th fog device is determined from Eqs. (10) and (11) as,

$$\begin{aligned} \mathbb{D}_{f_i}^{\text{que}} &= \frac{1}{n} \left((n-1)\mathbb{D}_{f_i}^{\text{proc}}(t_1) + (n-2)\mathbb{D}_{f_i}^{\text{proc}}(t_2) \right. \\ &\left. + (n-3)\mathbb{D}_{f_i}^{\text{proc}}(t_3) + \dots + \mathbb{D}_{f_i}^{\text{proc}}(t_{n-1}) \right) \end{aligned} \quad (12)$$

Finally, the total delay of fog device f_i is calculated from Eqs. (4), (6), (8), and (11) as

$$\mathbb{D}_{f_i} = \mathbb{D}_{f_i}^{\text{trans}} + \mathbb{D}_{f_i}^{\text{prop}} + \mathbb{D}_{f_i}^{\text{proc}} + \mathbb{D}_{f_i}^{\text{que}} \quad (13)$$

Definition 4. The fog residual energy ($\mathbb{R}$) is measured by the leftover energy in the fog device after the total energy consumed for EEG data processing (J^{proc}), data transmission (J^{trans}), receiving (J^{recv}), and the idle energy (J^{idle}).

Let the total energy of a fog device at time t be denoted as J^{total} , which is the battery capacity of the device. Let J^{proc} be the energy consumption by a fog device for processing of EEG data. Further, each fog device consumes some transmission energy (J^{trans}) and receiving energy (J^{recv}). For green computing, we prefer to convert the fog device energy consumption from normal operational state to low-power state when idle. However, fog devices consume a bit of energy in the idle state, denoted as (J^{idle}). Considering j_{f_i} as energy consumption of i th fog device per unit time, we have,

$$J_{f_i}^{\text{proc}} = j_{f_i} \times \frac{1}{c_{f_i}^{\text{max}}} \sum_{k=1}^n \mathcal{V}(t_k) \quad (14)$$

$$J_{f_i}^{\text{trans}} = J_{f_i}^{\text{recv}} = j_{f_i} \times \frac{1}{\mathcal{R}} \sum_{j=1}^E \sum_{k=1}^{T_j} \mathcal{V}(t_{jk}) \quad (15)$$

In Eq. (15), we consider the transmission energy the same as the receiving energy assuming equal bandwidth $\mathcal{B}$ for uploading and down-

loading channels. This symmetry provides balanced energy efficiency avoiding bias in selection process. From Eqs. (14) and (15), the residual energy ($\mathbb{R}_{f_i}$) for i th fog device is formed in Eq. (16) as,

$$\begin{aligned} \mathbb{R}_{f_i} &= J^{\text{total}} - \left[\left(j_{f_i} \times \frac{1}{c_{f_i}^{\text{max}}} \sum_{k=1}^n \mathcal{V}(t_k) \right) \right. \\ &\left. + \left(j_{f_i} \times \frac{2}{\mathcal{R}} \sum_{j=1}^E \sum_{k=1}^{T_j} \mathcal{V}(t_{jk}) \right) + J^{\text{idle}} \right] \end{aligned} \quad (16)$$

Definition 5. The fog memory efficiency ($\mathbb{M}_{f_i}$) of i th fog device is the balanced trade-off function between normalized primary memory ($M_{f_i}^{[P/\text{tot}]}$) and secondary memory ($M_{f_i}^{[S/\text{tot}]}$) associated with it. Where α denotes the weight of total primary memory ($M_{f_i}^{[P/\text{tot}]}$) and β denotes the weight of total secondary memory ($M_{f_i}^{[S/\text{tot}]}$).

During the task execution time, a task is being executed by a fog device occupying the primary memory of the device. While other assigned tasks are present in the secondary memory, waiting for their turn. Therefore, it is important to consider both memories in determining memory efficiency. We consider equal weightage and normalized values for both types of memories considering their free available spaces.

$$\mathbb{M}_{f_i} = \alpha \times \frac{M_{f_i}^{[P/\text{tot}]} - M_{f_i}^{[P/\text{occ}]}}{M_{f_i}^{[P/\text{tot}]}} + \beta \times \frac{M_{f_i}^{[S/\text{tot}]} - M_{f_i}^{[S/\text{occ}]}}{M_{f_i}^{[S/\text{tot}]}} \quad (17)$$

where $\alpha = \beta = 0.5$, and $M^{[*/\text{occ}]}$ indicates how much memory occupied in the device.

Definition 6. The fog performance efficiency ($\mathbb{P}_{f_i}$) of i th fog device is directly proportional to the million instructions per second ($\mathcal{M}_{f_i}$) and the number of cores (ζ_{f_i}), and reciprocal to the average clock frequency ($c_{f_i}^{\text{avg}}$).

The performance efficiency of a fog device depends on three parameters - (i) number of cores associated with the processor, (ii) execution capacity in a million instructions per second, and (iii) processor clock frequency. Therefore, $\mathbb{P}_{f_i}$ is calculated as

$$\mathbb{P}_{f_i} = \frac{\zeta_{f_i} \times \mathcal{M}_{f_i} \times 10^6}{c_{f_i}^{\text{avg}}} \quad (18)$$

3.2. Defining relations and functions

For $\mathcal{F} = \{f_1, f_2, f_3, \dots, f_F\}$, consider an element $\bar{f}_i$ from $\mathcal{F}$ represented as $\bar{f}_i \in \mathcal{F}'$, where $\mathcal{F} \cup \mathcal{F}' = \mathcal{F}$ and $\mathcal{F} \cap \mathcal{F}' = \mathcal{F}'$. Let $\mathcal{C}$ be the set of criteria represented as $\mathcal{C} = \{C_1, C_2, C_3, \dots, C_m\}$ and $|\mathcal{C}| = m$. Let $f_i \rightarrow C_a$ represents the evaluation of f_i on criteria C_a . For a pair of fog devices $\langle f_i, \bar{f}_i \rangle$ evaluating on a single criterion C_a is denoted as a function $\mathcal{P}(f_i > \bar{f}_i) \rightarrow C_a$, where $f_i > \bar{f}_i$ is the relation between two fog devices, indicating f_i is preferred over $\bar{f}_i$. Again, the aggregated function $\mathcal{P}^{\boxplus}(f_i > \bar{f}_i)$ for each pair of fog devices evaluated on each criterion is represented as

$$\mathcal{P}^{\boxplus}(f_i > \bar{f}_i) = \frac{\sum_{a=1}^m w_a \cdot \mathcal{P}(f_i > \bar{f}_i)}{\sum_{a=1}^m w_a} \quad (19)$$

where, w_a is the weight of C_a representing its relative importance. For each criterion, considering any pair of fog devices, the function $\mathcal{P}(f_i > \bar{f}_i) \rightarrow C_a$ is anti-symmetric:

$$\mathcal{P}(f_i > \bar{f}_i) = 1 - \mathcal{P}(\bar{f}_i > f_i) \quad (20)$$

3.3. Problem formulation

The main objective of E2F technique is the optimal selection of the master head among multiple fog devices, which is derived from the four factors- Fog Service Delay ($\mathbb{D}$), Fog Residual Energy ($\mathbb{R}$), Fog Memory Efficiency ($\mathbb{M}$), Fog Performance Efficiency ($\mathbb{P}$). The set of all four factors are denoted by $C = \{\mathbb{D}, \mathbb{R}, \mathbb{M}, \mathbb{P}\}$, where C_a denotes a criterion. Mathematically, our objective function ($\mathcal{O}(f)$) is formulated as:

$$\mathcal{O}(f) = \max_{f_i \in F} \left[\frac{1}{F-1} \left\{ \sum_{f_i \in F} \left(\sum_{a=1}^m w_a \cdot \mathcal{P}(f_i > \bar{f}_i) \rightarrow C_a \right. \right. \right. \\ \left. \left. \left. - \sum_{a=1}^m w_a \cdot \mathcal{P}(\bar{f}_i > f_i) \rightarrow C_a \right) \right\} \right] \quad (21)$$

subject to

$$\text{Condition 1 : } \sum_{a=1}^m w_a = 1$$

$$\text{Condition 2 : } 0 \leq w_a \leq 0.5$$

$$\text{Condition 3 : } 0 \leq \mathcal{P}(f_i > \bar{f}_i) \leq 1, \forall i = [1, F]$$

$$\text{Condition 4 : } \mathcal{P}(f_i > \bar{f}_i) = 0, \forall \mathcal{P}(f_i > \bar{f}_i) \leq 0$$

In Eq. (21), Condition 1 and 2 indicate that the criterion weights are nonzero and the sum of weights of all the criteria must be equal to 1, whereas Condition 2 limits weight for any single criterion to 0.5 or less. Condition 4 signifies the preference of one fog device over another is set to 0 if the preference value is negative. Moreover, Condition 2 and 3 ensure an unbiased and balanced comparison providing non-domination of any of the fog devices over others.

4. Proposed methodology

The proposed E2F architecture, shown in Fig. 1 consists of multiple EEG devices (E) in a single platform that are stand-alone and user-centric and configured with multiple channels (N). The signals are collected at the edge gateway in the form of data and transmitted to the fog layer. In the fog layer, we propose to select one of the designated fog devices, called the Master Head (MH), that receives the EEG data from the edge gateway and manages task distribution among fog devices. For optimal MH selection, we consider the key fog parameters such as clock frequencies ($c^{\min}, c^{\max}$), battery capacity (E), device memories ($S^{\text{pri}}, S^{\text{sec}}$), number of cores (ζ), and fog processor capacity ($\mathcal{M}$). Furthermore, we integrate the 6G features with the fog devices that are part of the IoT network. However, due to the limited storage and processing capacity of the fog devices, the MH is connected to the cloud layer for permanent storage of any extensive processing of EEG data. Fig. 2 depicts the workflow of the proposed work.

4.1. Proposed master head selection

The master head selection technique involves four factors- $\mathbb{D}$, $\mathbb{R}$, $\mathbb{M}$, and $\mathbb{P}$ from each of the fog devices. Considering each factor as a criterion of the devices, we transform our problem into a correlation-weighted multi-criteria decision-making approach. Among the criteria, $\mathbb{R}$, $\mathbb{M}$, and $\mathbb{P}$ are related proportionately to the preference of the master head selection. However, $\mathbb{D}$ is inversely proportionate, meaning the lesser the fog delay factor, the higher the preference of the fog device to be the master node. Therefore, we aim to select an optimal fog device that minimizes the fog service delay and maximizes the other factors. The objective function is reformulated as a matrix-vector representation for f_i as,

$$\mathcal{O}(f) = \max_{f_i \in F} \left(\frac{1}{F-1} (\mathbb{M}_i - \mathbb{M}_i^T) \bar{w} \right) \quad (22)$$

where, $\mathbb{M}_i$ is the matrix for i th fog device following $f_i \rightarrow C_a$, that is an $F \times m$ criteria-preference matrix considering all pairwise preference values of fog device f_i over every other fog devices across all criteria. $\mathbb{M}_i^T$ is transpose of $\mathbb{M}_i$, and $\bar{w}$ is the weight vector for C .

Theorem 1. *Selecting the optimal fog device, called as the master head, among F fog nodes considering m criteria is an NP-hard problem.*

Proof. We prove this by comparing our problem with the Multi-Dimensional Knapsack Problem (MDKP) defined in [36,37]. The MDKP problem is introduced as follows — let n be the number of items, and each item is associated with m attributes, where a_{ij} represents the value of j th attribute for i th item. Furthermore, c_j denotes the total capacity of j th attribute and b_i denotes the total benefit of i th item. The objective of MDKP is to select the items with the maximum possible benefits, defined as $\max \sum_{i=1}^n b_i s_i$, subject to $\sum_{i=1}^n a_{ij} s_{ij} \leq c_j, \forall j$. The constraint indicates that the sum of the maximum value of selected items must not exceed the total capacity. Further, $s_i \in \{0, 1\}$ indicated the selection criteria of i th item, which is an NP-hard problem. If item i is selected, then $s_i = 1$, otherwise $s_i = 0$. Now, we connect our problem with MDKP as follows — let F be the number of fog devices evaluated across m criteria is same as the n and m for MDKP, respectively. Let g_{ij} be the preference value w.r.t $f_i \rightarrow C_j$ similar to a_{ij} in MDKP. The modified master node selection problem can be defined as $\max \sum_{i=1}^F \sum_{j=1}^m w_j g_{ij}$, subject to $g_{ij} \geq \mathcal{T}_j, \forall i, j$, where $\mathcal{T}_j$ is preference threshold of criterion j . Both MDKP and the Master Head Selection Problem are NP-hard due to their combinatorial nature and the need to evaluate and compare items/fog devices across multiple dimensions/criteria. This comparison validates that the Master Head Selection Problem using multiple criteria is NP-hard, drawing a strong analogy to the computational complexity and challenges posed by the MDKP.

The solution to the objective function $\mathcal{O}(f)$ involves a weighted multiple criteria-based decision-making approach applied to each fog device on four criteria: fog delay efficiency, fog residual energy, fog memory efficiency, and fog performance efficiency, for selecting an optimal master head. Algorithm 1 computes relative weight of each decision criterion using a combination of normalization, statistical variability, and inter-criteria correlation. Initially, the raw criterion values have different magnitudes and ranges of values compared to one another. Therefore, the criteria are normalized to ensure comparability. Next, we follow the method of criteria importance through an inter-criteria correlation approach. This method combines two aspects- (i) the standard deviation of each criterion is computed to quantify its discriminative across fog devices, and (ii) the Pearson correlation is calculated for every pair of criteria to avoid assigning excessive importance to redundant criteria. Finally, for a given C_a , a composite importance score is obtained by multiplying its standard deviation with the sum of redundancy-penalization factor over all other criteria. These scores are then normalized to produce preliminary weights whose sum equals one. Moreover, Algorithm 2 shows the master head selection approach for E2F architecture. The initial process follows two variants of min-max normalization, based on the criteria values as maximization or minimization as shown in Eqs. (23) and (24), respectively.

$$\phi'_{ia} = \frac{\phi_{ia} - \min(\phi_{ia})}{\max(\phi_{ia}) - \min(\phi_{ia})} \quad (23)$$

$$\phi'_{ia} = \frac{\max(\phi_{ia}) - \phi_{ia}}{\max(\phi_{ia}) - \min(\phi_{ia})} \quad (24)$$

In Eqs. (23) and (24), $\phi_{ia} = f_i \rightarrow C_a, i = \{1, 2, 3, \dots, F\}$ and $a = \{1, 2, 3, \dots, m\}$. ϕ'_{ia} is the normalized value of ϕ_{ia} . The next step includes the calculation of preference values of fog devices over the given criteria. The preference values are calculated as follows

$$\mathcal{P}(f_i > \bar{f}_i) \rightarrow C_a = \begin{cases} 0 & \text{if } \phi'_{ia} - \phi'_{ia} \leq 0 \\ \phi'_{ia} - \phi'_{ia} & \text{if } \phi'_{ia} - \phi'_{ia} > 0 \end{cases} \quad (25)$$

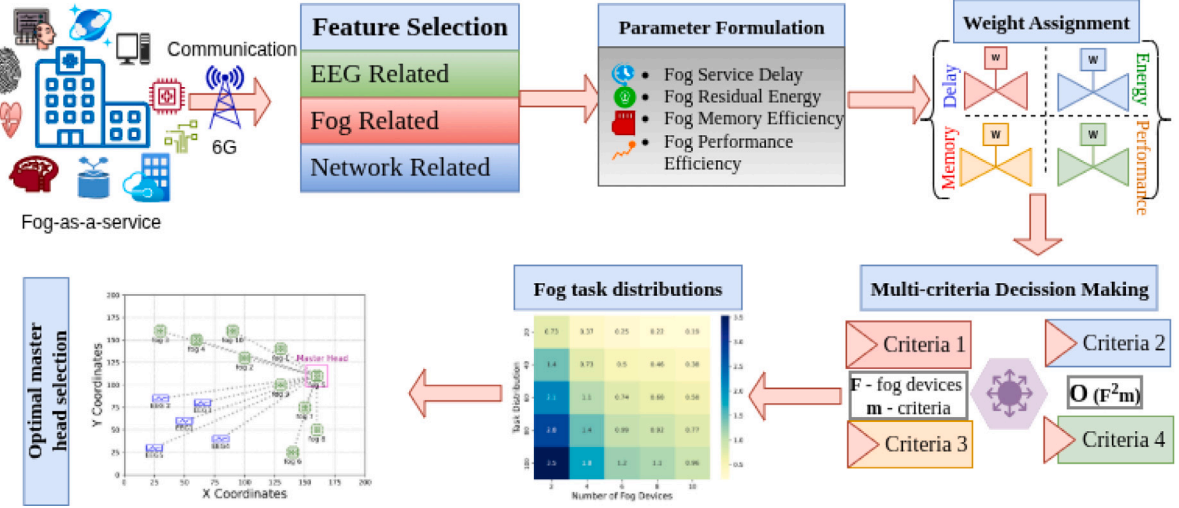


Fig. 2. Shows the workflow of the proposed work.

Algorithm 1: Criterion Weight Calculation

Input: Decision matrix $\Phi = [\phi_{ia}]$, Criteria $C = \{\mathbb{D}, \mathbb{R}, \mathbb{M}, \mathbb{P}\}$
Output: Weight vector $\vec{w}$

- 1 Normalize each criterion;
- 2 Calculate mean $\bar{\phi}'_a = \frac{1}{F} \sum_{i=1}^F \phi'_{ia}$;
- 3 **for each criterion** $C_a \in C$ **do**
- 4 Compute standard deviation $\sigma_a = \sqrt{\frac{1}{F} \sum_{i=1}^F (\phi'_{ia} - \bar{\phi}'_a)^2}$;
- 5 **for each** $C_{b \neq a} \in C$ **do**
- 6 Compute $r_{ab} = \frac{1}{F \sigma_a \sigma_b} \sum_{i=1}^F (\phi'_{ia} - \bar{\phi}'_a)(\phi'_{ib} - \bar{\phi}'_b)$;
- 7 $C_a \leftarrow \sigma_a \sum_{b \neq a} (1 - r_{ab})$;
- 8 $w_a = \frac{C_a}{\sum_{k=1}^{|C|} C_k}$;
- 9 **return** w_a ;
- 10 **return** $\vec{w}$;

where, $f_i \in F$, $\bar{f}_i \in F'$, and $F - F' = \{f_i\}$. For example, two fog devices f_1 and f_2 being compared on a single normalized criterion C_1 . Let, their normalized values are 0.80 and 0.65 respectively. Using Eq. (25), the pairwise preference $\mathcal{P}(f_1 > f_2) \rightarrow C_1 = 0.15$. This means that f_1 is preferred over f_2 by a strength of 0.15 on criterion C_1 . If instead $\phi'_{11} \leq \phi'_{21}$, then the preference value would be zero, indicating no advantage. Next, the aggregated function is calculated using Eq. (19), satisfying Condition 1 of $\mathcal{O}(f)$ as

$$\mathcal{P}^{\boxplus}(f_i > \bar{f}_i) = \sum_{a=1}^m w_a \cdot \mathcal{P}(f_i > \bar{f}_i) \quad (26)$$

The resultant of $\mathcal{P}^{\boxplus}(f_i > \bar{f}_i)$ forms a square matrix containing the mapped values from f_i to $\bar{f}_i$ as follows:

$$\mathbb{M}^{F \times F} = \begin{pmatrix} \mathcal{P}^{\boxplus}(f_1 > \bar{f}_1) & \mathcal{P}^{\boxplus}(f_1 > \bar{f}_2) & \dots & \mathcal{P}^{\boxplus}(f_1 > \bar{f}_F) \\ \mathcal{P}^{\boxplus}(f_2 > \bar{f}_1) & \mathcal{P}^{\boxplus}(f_2 > \bar{f}_2) & \dots & \mathcal{P}^{\boxplus}(f_2 > \bar{f}_F) \\ \vdots & \vdots & \ddots & \vdots \\ \mathcal{P}^{\boxplus}(f_F > \bar{f}_1) & \mathcal{P}^{\boxplus}(f_F > \bar{f}_2) & \dots & \mathcal{P}^{\boxplus}(f_F > \bar{f}_F) \end{pmatrix}$$

The matrix $\mathbb{M}^{F \times F}$ is used to find the net outranking flow $\varphi(f_*)$ for each fog device. The net outranking flow $\varphi(f_i)$ of i th fog device is calculated as the difference between positive outranking flow $\varphi^+(f_i)$ and the negative outranking flow $\varphi^-(f_i)$ of the device. From $\mathbb{M}^{F \times F}$ matrix, the positive outranking flow is calculated using Eq. (27) as

$$\varphi^+(f_i) = \frac{1}{F-1} \left[\sum_{i=1}^F \mathcal{P}(f_i > \bar{f}_i) \right], f_i \neq \bar{f}_i \quad (27)$$

Algorithm 2: Master Head Selection

Input: Fog devices $F = \{f_1, f_2, \dots, f_F\}$, EEG devices $\mathcal{E} = \{e_1, e_2, \dots, e_E\}$, weight vector $\vec{w} = \{w_{\mathbb{D}}, w_{\mathbb{R}}, w_{\mathbb{M}}, w_{\mathbb{P}}\}$
Output: Optimal selection of master head among fog devices

- 1 Initialize decision matrix $\mathbb{D} = [\phi_{ia}]$ with $\phi_{ia} = (f_i \rightarrow C_a)$;
- 2 **for each** $f_i \in F$ **do**
- 3 **if** $C_a \in \{\mathbb{R}_{f_i}, \mathbb{M}_{f_i}, \mathbb{P}_{f_i}\}$ **then**
- 4 Update $\phi_{ia} \leftarrow \frac{\phi_{ia} - \min(\phi_{ia})}{\max(\phi_{ia}) - \min(\phi_{ia})}$;
- 5 **else**
- 6 Update $\phi_{ia} \leftarrow \frac{\max(\phi_{ia}) - \phi_{ia}}{\max(\phi_{ia}) - \min(\phi_{ia})}$;
- 7 $(w_{\mathbb{D}}, w_{\mathbb{R}}, w_{\mathbb{M}}, w_{\mathbb{P}}) \leftarrow \text{CalculateWeights}(\mathbb{D}, \mathbb{R}, \mathbb{M}, \mathbb{P})$;
- 8 **for each pair** $(f_i, \bar{f}_i)$ **with** $f_i \neq \bar{f}_i$ **do**
- 9 **for each** $C_a \in C$ **do**
- 10 **if** $\phi'_{ia} - \phi'_{i'a} \leq 0$ **then**
- 11 Assign $\mathcal{P}(f_i > \bar{f}_i) \leftarrow 0$;
- 12 **else**
- 13 Assign $\mathcal{P}(f_i > \bar{f}_i) \leftarrow (\phi'_{ia} - \phi'_{i'a})$;
- 14 $\mathcal{P}^{\boxplus}(f_i > \bar{f}_i) \leftarrow w_a \cdot \mathcal{P}(f_i > \bar{f}_i)$;
- 15 Construct $\mathbb{M}^{F \times F}$ w.r.t. $\mathcal{P}^{\boxplus}(f_i > \bar{f}_i)$;
- 16 $\varphi^+(f_i) \leftarrow \frac{1}{F-1} \sum_{i=1}^F \mathcal{P}(f_i > \bar{f}_i)$;
- 17 $\varphi^-(f_i) \leftarrow \frac{1}{F-1} \sum_{i=1}^F \mathcal{P}(\bar{f}_i > f_i)$;
- 18 $\varphi(f_i) \leftarrow \varphi^+(f_i) - \varphi^-(f_i)$;
- 19 **return** $\text{Masterhead} \leftarrow \arg \max_i \varphi(f_i)$;

Similarly, the negative outranking flow is calculated using Eq. (28) as

$$\varphi^-(f_i) = \frac{1}{F-1} \left[\sum_{i=1}^F \mathcal{P}(\bar{f}_i > f_i) \right], f_i \neq \bar{f}_i \quad (28)$$

Finally, $\varphi(f_i)$ is derived from Eqs. (27) and (28) as

$$\varphi(f_i) = \varphi^+(f_i) - \varphi^-(f_i) \quad (29)$$

We calculate the net outranking flow for each fog device using Eq. (29). We select the highest value of the fog device as the master head and the second highest value as the alternate master head.

Theorem 2. The computational complexity of the proposed master head selection is $\mathcal{O}(F^2m)$.

Table 2
Synthetic fog dataset with fog-related parameters.

Fog parameters	$c^{\min}$ (GHz)	$c^{\max}$ (GHz)	ζ	$\mathcal{M}$	S^{pri} (GB)	S^{sec} (GB)	E (Joules)
Fog 1	2.6	3.5	2	24 000	4	128	234 000
Fog 2	2.6	3.8	4	20 000	6	64	198 000
Fog 3	1.8	3.2	6	15 000	6	256	162 000
Fog 4	2.4	4.2	4	12 080	8	128	133 200
Fog 5	3.0	4.1	2	19 200	6	128	216 000
Fog 6	2.5	3.4	2	20 800	6	64	144 000
Fog 7	1.5	1.5	4	12 000	8	64	234 000
Fog 8	1.4	1.4	4	11 200	4	256	162 000
Fog 9	1.8	2.8	8	15 000	4	128	133 200
Fog 10	1.4	1.9	6	26 800	4	64	162 000

Proof. The computational complexity of our proposed technique is mainly dependent on two entities - number of fog devices (F) and number of criteria (m). For F fog devices, we have $F \times (F-1)/2$ pairwise comparisons, assuming all devices are compared against each other. Each comparison involves evaluating m criteria, so the complexity for all comparisons is $O(F^2m)$. Further, the calculation of net outranking from positive and negative outranking flows takes $O(F^2m)$. Summing preference indices for each fog device involves iterating over F fog devices and m criteria. The last step of sorting the fog devices based on net outranking is $O(F \log F)$. Combining all the steps, the overall complexity of the master head selection is $O(F^2m)$. However, the variations in the computational complexity can be observed based on the relation between F and m as:

- Case 1: If $F \gg m$, then computational complexity is $O(F^2m)$. This helps in generating more accurate device preferences, leading to select the best device.
- Case 2: If $m \gg F$, then computational complexity is $O(m^3)$. It evaluates more comprehensive criteria, resulting in more accurate weight distribution.
- Case 3: If $F \simeq m$, then computational complexity is $O(m^2F) + O(F^2m) \approx O(F^2m)$. This is a balanced evaluation approach and gives moderate scalability.

4.2. Illustrative example

Let us consider the scenario where $F = 10$, $m = 4$, $E = 5$, and $T = 100$. The fog-related parameters are shown in Table 2. The table shows the specifications of each fog device selected in such a way that no single fog device dominates other fog devices in the network. We calculate the step-by-step process of the master head selection among the fog devices as follows.

In first step, We calculate the four criteria- $\mathbb{D}, \mathbb{R}, \mathbb{M}, \mathbb{P}$ derived from Eqs. (13), (16), (17), and (18) respectively for each fog device given in Table 3. In step-2, the respective weights $w_{\mathbb{D}}, w_{\mathbb{R}}, w_{\mathbb{M}}, w_{\mathbb{P}}$ are calculated in Eq. (30) following Algorithm 1. The sum of total weights of all the criteria equals to 1, as given in Condition 1 and 2 of Eq. (21). In this example, fog residual energy and fog performance efficiency carry equal weight.

$$\vec{w} = \{0.28, 0.25, 0.22, 0.25\} \quad (30)$$

In the next step, We apply Eq. (26) to compute the weighted preference value on the normalized values of given Table 3. Next, the resultant square matrix $\mathbb{M}^{10 \times 10}$ is formed (see Box I). Finally, the last step calculates φ^+ , using Eq. (27) and φ^- , using Eq. (28) for fog devices 1 to 10 to derive the net outranking flow ($\varphi(f)$), using Eq. (29) as

$$\varphi(f) = [0.1659, 0.0779, 0.0713, -0.0467, \mathbf{0.1896}, -0.1187, 0.0166, -0.1177, -0.1335, -0.1046] \quad (31)$$

From Eq. (31), we observe that f_5 has the highest net outranking value out of all the fog devices. Therefore, in this scenario, f_5 is selected as the master head.

In Fig. 3, scenario-1, the EEG and fog devices operate within a distributed environment. We observe that the selected master head,

Table 3
Fog criteria evaluation.

Fog ID	Fog criteria			
	$\mathbb{D}$ (in ms)	$\mathbb{R}$ (in Joules)	$\mathbb{M}$	$\mathbb{P}$
Fog 1	56.9062	233 993.0056	0.0857	15.7377
Fog 2	56.8994	197 994.0817	0.0786	25.0000
Fog 3	60.6151	161 994.9458	0.1536	36.0000
Fog 4	57.5801	133 195.9859	0.1214	14.6424
Fog 5	55.7913	215 993.6284	0.1036	10.8169
Fog 6	57.2421	143 995.6788	0.0786	14.1017
Fog 7	62.9986	233 992.5004	0.0964	32.0000
Fog 8	64.0429	161 994.7489	0.1357	32.0000
Fog 9	60.5823	133 195.8443	0.0857	52.1739
Fog 10	64.0253	161 994.7489	0.0607	97.4545

f_5 , has no advantage of having the highest magnitude of parameter values, but the optimal selection is done based on the criteria values. The $\mathbb{E}2\mathbb{F}$ technique is proven to be effective in satisfying minimum $\mathbb{D}$ value and maximum $\mathbb{R}, \mathbb{M}, \mathbb{P}$ values. Similarly, in scenario-2, the EEG-fog placement is within a clustered environment. Considering the same parameter values for the fog devices given in Table 2. We see that the same f_5 is selected as the master head among all fog devices.

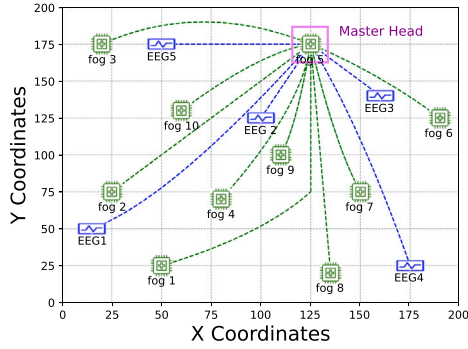
5. Experimental analysis

This section exhibits the results of $\mathbb{E}2\mathbb{F}$ technique and highlights the performance of the work considering different parameters. We follow the M/M/1 system model as the standard, assuming there is one healthcare server and each fog device has a single queue. We consider the communication protocol and the medium of communication to be IEEE 802.11 and Bluetooth Low Energy (BLE), respectively. Further, we analyze the effectiveness of this work compared to existing techniques - 3D-POS [38], AOA [39], GA-COA [40], KCHE [41], AMO [42], OFA [43], Indranil et al. [44], and MHHO [45].

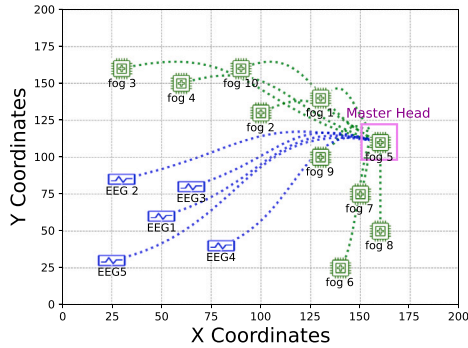
- 3D-POS - A Pareto optimization based fog head selection in multi-fog healthcare IoT network for EEG applications.
- AOA - A dynamic load balancing approach in fog-cloud environment.
- GA-COA - An IoT based multi-fog network for optimal resource utilization.
- KCHE - A lossless EEG data compression technique with epileptic seizure prediction using edge-fog computing.
- AMO - An energy-aware task scheduling model aiming to reduce total system delay and energy consumption using fog computing platform.
- OFA and Indranil et al. - Effective task offloading strategies among fog devices based on their processing capabilities.
- MHHO - A task distribution strategy between fog and cloud layer, considering makespan time of the network.

$$M^{10 \times 10} = \begin{pmatrix} - & 0.1062 & 0.3042 & 0.2767 & 0.0588 & 0.2562 & 0.2073 & 0.4209 & 0.3756 & 0.4801 \\ 0.0267 & - & 0.2149 & 0.2142 & 0.0409 & 0.1767 & 0.2073 & 0.3316 & 0.2863 & 0.3739 \\ 0.2192 & 0.2094 & - & 0.2092 & 0.1911 & 0.2855 & 0.2286 & 0.1705 & 0.2322 & 0.3367 \\ 0.0846 & 0.1015 & 0.1021 & - & 0.0533 & 0.1031 & 0.2430 & 0.2188 & 0.1867 & 0.3626 \\ 0.0800 & 0.1416 & 0.2973 & 0.2666 & - & 0.2868 & 0.2619 & 0.4139 & 0.4110 & 0.5155 \\ 0 & 0 & 0.1143 & 0.0390 & 0.0095 & - & 0.1960 & 0.2310 & 0.1411 & 0.2733 \\ 0.0723 & 0.1518 & 0.1786 & 0.3001 & 0.1058 & 0.3172 & - & 0.2136 & 0.2754 & 0.2982 \\ 0.1654 & 0.1556 & 0 & 0.1554 & 0.1373 & 0.2317 & 0.0931 & - & 0.1899 & 0.1777 \\ 0.1051 & 0.0953 & 0.0467 & 0.1083 & 0.1193 & 0.1268 & 0.1399 & 0.1749 & - & 0.1759 \\ 0.2358 & 0.2091 & 0.1773 & 0.3104 & 0.2500 & 0.2852 & 0.1889 & 0.1889 & 0.2021 & - \end{pmatrix}$$

Box 1.



(a)



(b)

Fig. 3. EEG-fog placement with Master head selection: (a) *scenario-1*: Shows the configuration of EEG and fog devices in a distributed environment, highlighting the spatial distribution and connectivity of devices. and (b) *scenario-2*: Illustrates the EEG-fog placement in a clustered environment, where devices are organized in a more localized grouping.

5.1. Simulation setup

For our work, we consider the number of fog devices and the number of EEG devices as $\mathcal{F} = [2, 10]$ and $\mathcal{E} = [1, 5]$, respectively. The total number of tasks is considered as $\mathcal{T} = [1, 100]$. We include 6G-aware parameter values in our $\mathbb{E}2\mathbb{F}$ architecture for the simulation area of 200×200 square unit. According to the 6G characteristics, the parameters are considered as $c^{\text{avg}} = [1.4, 3.5]$ GHz and small scale channel bandwidth $B = 30$ MHz [35]. Furthermore, the network parameters are considered as minimal SNR = [20, 30] dB and $\mathcal{M} = [11200, 26800]$. We set the maximum value for $M^{[P/\text{tot}]}$ and $M^{[S/\text{tot}]}$ as 8 GB and 256 GB respectively. Other fog simulation parameter values

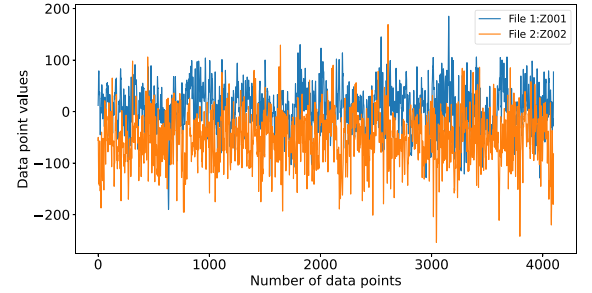


Fig. 4. Distribution of EEG data points and their correlation across the number of channels, based on two files from the Z subject of the Bonn University EEG dataset.

are referred from Table 2. We perform 100 iterations and consider the average value for each result analysis.

5.2. EEG dataset description

The EEG characteristics are taken from the Bonn University dataset [46]. The dataset contains five different subjects, named as- Z, N, F, O, and S. Each subject consists of 100 text files and each file has 4097 data points. The EEG recordings are done with a sampling frequency of 173.61 Hz from 100 channels. For our experiment, we considered all the files of Z subject as the number of tasks. Fig. 4 represents EEG data points of the first 2 files of Z subject.

5.3. Performance analysis

We evaluate the performance of our work by examining various parameters, including fog service delay and fog residual energy. Initially, we focus on transmission delay as a performance metric, considering two different network bandwidths: the standard 20 MHz and the 6G 30 MHz, based on the given signal-to-noise ratio [35], and the channel capacity $\mathcal{R}$ for the given bandwidth B , while varying the number of tasks. Our results show that the average transmission delay decreases from 0.0908 to 0.0605 mSec across 100 tasks. Fig. 5 illustrates the variation in transmission delay for 20, 40, 60, 80, and 100 tasks.

We examine the propagation delay by considering the distance between EEG devices and fog devices with a fixed data propagation speed. Fig. 6 illustrates the data travel time between EEG devices and fog devices, as well as the travel time between EEG devices and the cloud. Our findings indicate that the propagation delay increases with higher workloads and fewer fog nodes. For a single EEG device, the average propagation delay ranges from 0.1193 mSec with 2 fogs to 0.0243 mSec with 10 fogs. while the delay is significantly higher for the cloud that is 0.417 mSec. This trend is consistent with an increasing number of EEGs.

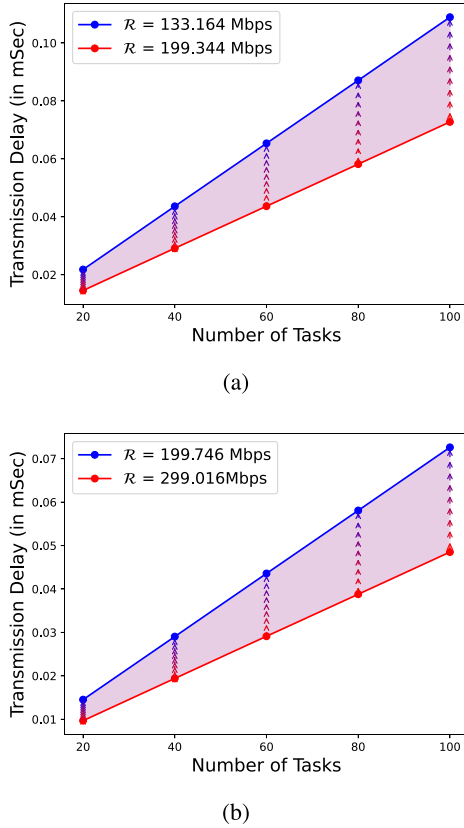


Fig. 5. EEG data transmission time across several tasks (a) over a standard healthcare IoT network with SNR as 20–30 dB, and (b) for 6G over the healthcare IoT network under the same SNR. The blue lines represent the maximum delay, while the red line indicates the minimum delay.

The processing delay is influenced by both the volume of EEG data and the clock frequency of the fog devices. Therefore we analyze the processing delay by varying the clock frequency of the fog devices across multiple tasks. We consider the average clock frequency of 6G-enabled fog devices, $c^{avg} = 2.4$ GHz to calculate the processing time. We observe that the average processing time for a fog device to execute 100 tasks is 10.66 mSec operating on below c^{avg} , which is twice when operating above c^{avg} , that is 5.69 mSec. Additionally, we add the average queuing delay to calculate the total computation time which increases with the number of tasks, ranging from 2.92 mSec to 15.61 mSec. The change in total processing delay and total computation delay across multiple tasks are depicted in Fig. 7.

Fig. 8 illustrates the fog service delay, representing the total average delay incurred across multiple fog executions for a given task distribution. It is observed that as the number of fogs increases, the service delay decreases but is inversely proportional to the task count. In our analysis, the minimum and maximum service delays are [0.192, 0.729] mSec for 20 tasks executed in 10 fog devices. However, with an increase to 100 tasks, the range widens to [0.959, 3.534] mSec.

We analyze the total energy consumption by the fog devices for the number of tasks and observe that the energy consumption for 20 tasks is 90.6 to 123.7 Joules for 2–10 fog devices. For the same fog devices, it is 452.51 to 618.05 Joules considering 100 tasks. This is because more tasks consume higher transmission and receiving energy and takes more processing time. Fig. 9 shows the detailed energy consumption by a number of fog devices with an increased number of tasks.

Additionally, we evaluate three essential performance indicators — energy consumption per task, system throughput, and effects of MH selection on resource utilization. We evaluate energy-per-task on fog

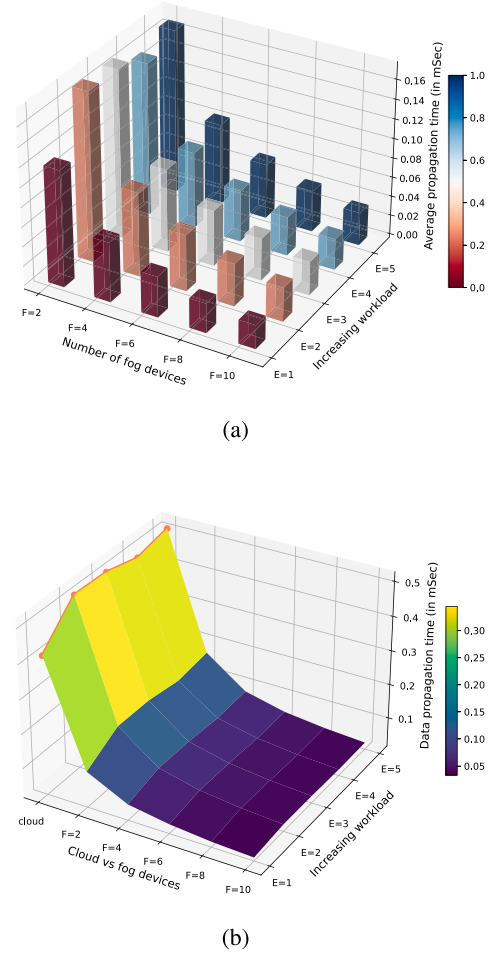
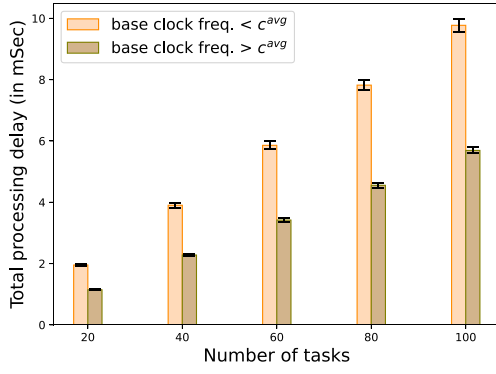


Fig. 6. Average data propagation time between EEG devices and fog devices in E2F architecture. (a) shows the variation in average propagation delay as the number of fog devices and workload increase, and (b) shows the comparison of average propagation delay between fog devices and the cloud.

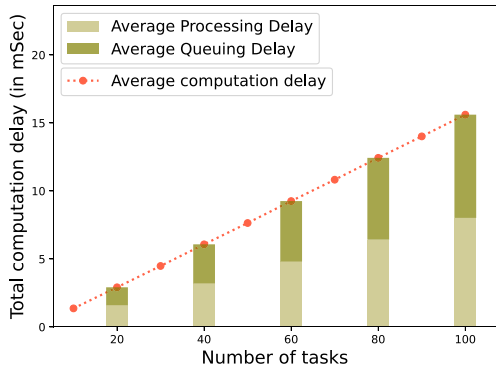
devices with c^{min} that is the sum of processing energy, transmission and receiving energy, along with idle energy at fog device per task. We observe the consumption energy per task is 4.52J for minimum fog configuration with energy-balanced fog operations. The system throughput is evaluated as the total tasks processed in required computation time. Our proposed system achieves a higher throughput of approximately 64 tasks per second compared to random distribution of 28 tasks per second. The MH performs intelligent allocation based on residual energy, delay, and memory, resulting in near-uniform resource utilization across fog nodes. Without MH selection, CPU and memory utilization show high variance (up to 52%) and with MH selection, variance drops to 14–18%.

5.4. Comparative analysis

To ensure a fair comparison, we adopt the execution environments and parameter settings explicitly reported in the original studies. To ensure comparability, we have unified normalized in units and scale while retaining the original results. We evaluate average execution time with a fixed set of 100 tasks across all baselines. Kaushal et al. [38] have achieved an average execution time of 0.018 s under a high-bandwidth (500 Mbps) and high-frequency (average 3.0 GHz) environment for Pareto-based fog-head selection. but this work does not incorporate relative weighting or inter-criteria importance among fog parameters, which limits its decision robustness for highly scalable environment.



(a)

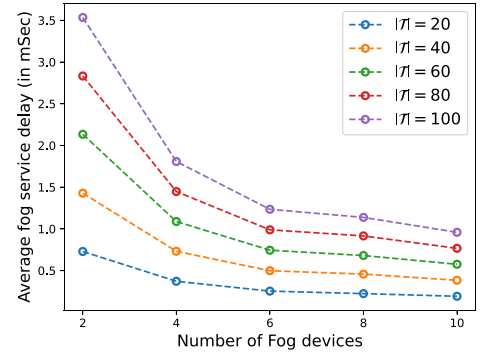


(b)

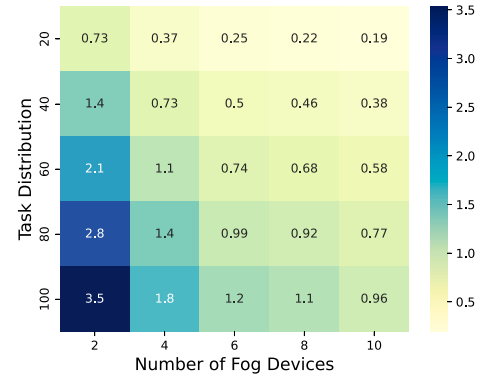
Fig. 7. Analysis of EEG data processing and computation times for a set of tasks ranging from 20 to 100. (a) shows variation in processing delay with changes in the clock frequencies of fog devices, considering the threshold as the average clock frequency, and (b) shows computation delay including the average queuing delay for the same number of tasks.

Navid et al. [39] have performed dynamic load balancing in a fog-cloud environment with a 2.5 GHz processing frequency, 5000-unit fog processing capability, and 50 simulation iterations. This work produces a significantly higher execution time of 1.341 s, reflecting the overhead latency in the fog layer. Tripathy et al. [40] have assessed a multi-fog optimal resource-utilization strategy using an average fog clock rate of 2.6 GHz with 1500 processing capability and variable task sets ranging from 100 to 800 tasks. This strategy achieves an average latency of 0.046 s with 50 iterations. In contrast, our proposed method achieves substantially lower average execution time of 0.0156 s for the same 100-task workload. It shows, $\mathbb{E}2\mathbb{F}$ achieves 13–98% execution time reduction compared to the recent multi-fog IoT approaches.

Furthermore, the proposed work is compared with the studies by KCHE [41], which utilize a traditional fog computing IoT network for EEG data reduction and classification. The authors have calculated the total delay, including transmission time, link latency, waiting time, and processing time using the Bonn University EEG dataset, as considered in our work. The results highlight that the authors have reduced the total delay from 30 secs to an approximation delay of 2.5 secs using EEG data compression technique for Z subject. However, for the same scenario, our technique achieves a significant average delay of 59.67 μ secs. Considering 10 fog devices in a network, AMO [42] produces a latency of approximately 5 secs while task scheduling and OFA [43] produces a latency of approximately 1.5 secs while task offloading respectively. Whereas the computational offloading using heterogeneous fog devices in [44] scores a delay of 0.02 secs. Our proposed $\mathbb{E}2\mathbb{F}$ technique achieves a total latency of 596.68 μ sec for same number of fog devices.



(a)



(b)

Fig. 8. Analysis of service delay for $\mathbb{E}2\mathbb{F}$ architecture. (a) shows total service delay for multiple tasks (ranging from 20 to 100) executed on a number of fog devices (from 2 to 10), and (b) shows delay reference values for distinct numbers of fog devices over task distributions.

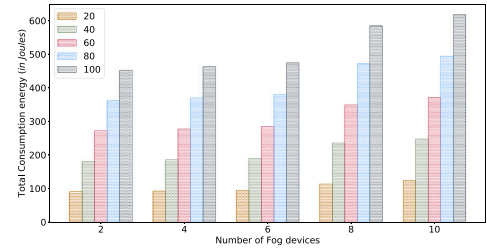


Fig. 9. Total energy consumption by fog devices (ranging from 2 to 10) with increasing workload (ranging from 20 to 100 tasks).

Similarly, for 100 tasks, MHHO [45] achieves the average makespan of approximately 7 millisecond executed on 10 devices in an IoT-fog-cloud computing environment. The results highlight that our proposed work suppresses all of the existing techniques by reducing the delay of 91%–99%. This significant reduction in delay is achieved due to the following reasons — the baseline methods inherently operate under conventional IoMT fog environments with low transmission bandwidth, and serial data processing in single fog device, increasing queuing and processing delays. In contrast, our proposed method is a 6G-driven FaaS IoT network with 6G supported network parameters and multiple collaborative fogs and MH. Fig. 10 exhibits the average execution time and total delay for different existing methods compared with our proposed $\mathbb{E}2\mathbb{F}$ method utilized fog computing services.

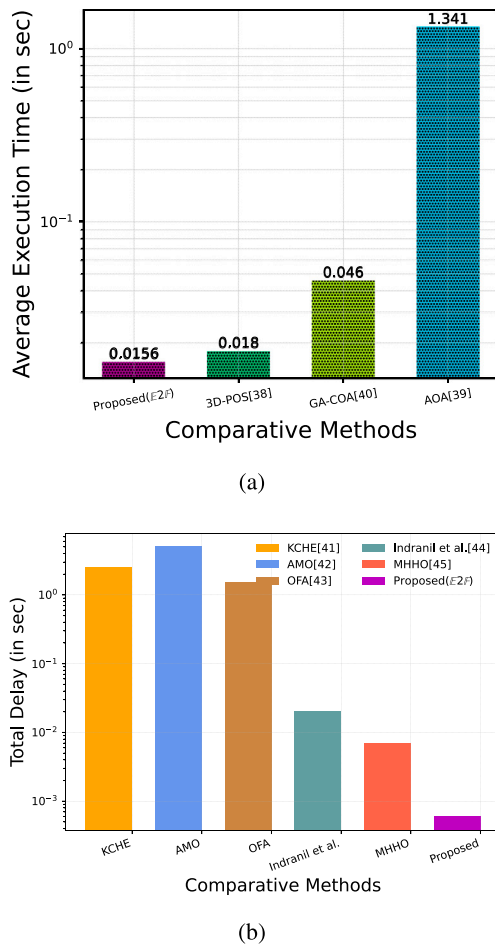


Fig. 10. Analysis of existing techniques using fog computing architecture compared with our proposed E2F technique. (a) For average execution time, (b) For total delay.

6. Conclusion

This work demonstrates a master head selection process for multi EEG multi fog IoT network using 6G parameterized fog devices. The primary objective is the selection of an optimal fog device among the fog devices that participate in the healthcare IoT network based on their parameter values. We consider the major parameters to evaluate multiple criteria that play a critical role in decision-making for the master head selection. Furthermore, we calculate respective criteria weights considering both the standard deviation of a single criterion and the correlation between a pair of criteria to increase the efficiency of the criteria weights. Though our proposed architecture is new for multiple EEG devices on a single platform, therefore we model our architecture and calculate various delays and energy factors with the change in the number of tasks and fog devices. The extensive experimental results show that 6G-enabled parameters in our fog network reduce 94% overall propagation delay compared to the cloud and more than 91% in total delay compared to the existing techniques. Although the present work primarily focuses on delay reduction and master-head selection within a collaborative fog architecture, the integration of advanced security and privacy mechanisms is an important direction. In the future direction, we aim to explore secure inter-fog communication and plan to implement light-weight encryption with trust scoring of fog nodes, and anomaly detection coordinated by the master head. Additionally, we focus on developing a novel EEG data compression technique to further minimize the service delay and an effective task scheduling technique for our E2F architecture.

CRedit authorship contribution statement

Rupalin Nanda: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Sakthivel P.:** Validation, Supervision. **Rama Krushna Rath:** Writing – review & editing, Visualization, Validation. **Abhishek Hazra:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] Rajesh Gupta, Dakshita Reebadiya, Sudeep Tanwar, 6G-enabled edge intelligence for ultra -reliable low latency applications: Vision and mission, *Comput. Stand. Interfaces* 77 (2021) 103521.
- [2] Nabajyoti Mazumdar, Amitava Nag, Jyoti Prakash Singh, Trust-based load-offloading protocol to reduce service delays in fog-computing-empowered iot, *Comput. Electr. Eng.* 93 (2021) 107223.
- [3] Theodore Sizer, Dragan Samardzija, Harish Viswanathan, Son Thai Le, Sarvesh Bidkar, Pascal Dom, Ed. Harstead, Thomas Pfeiffer, Integrated solutions for deployment of 6G mobile networks, *J. Lightwave Technol.* 40 (2) (2021) 346–357.
- [4] Guangyi Liu, Yuhong Huang, Na Li, Jing Dong, Jing Jin, Qixing Wang, Nan Li, Vision, requirements and network architecture of 6G mobile network beyond 2030, *China Commun.* 17 (9) (2020) 92–104.
- [5] Syed Thouheed Ahmed, V. Vinoth Kumar, Krishna Kant Singh, Akansha Singh, V. Muthukumar, Deepa Gupta, 6 g enabled federated learning for secure iomt resource recommendation and propagation analysis, *Comput. Electr. Eng.* 102 (2022) 108210.
- [6] Aqeel Thamer Jawad, Rihab Maaloul, Lamia Chaari, A comprehensive survey on 6G and beyond: Enabling technologies, opportunities of machine learning and challenges, *Comput. Netw.* 237 (2023) 110085.
- [7] Salwa Othmen, Radhia Khdir, Aymen Belghith, Khaled Hamoud, Chahira Lhioui, Doaa Elmourssi, Enhancing iot-enabled healthcare applications by efficient cluster head and path selection using fuzzy logic and enhanced particle swarm optimization, *Int. J. Commun. Syst.* 38 (3) (2025) e6096.
- [8] Nagma Irfan, Shuchi Dave, Vimanyu Veer, Emerging applications of artificial intelligence in analyzing eeg signals for the healthcare sector, *Artif. Intell. Biomed. Mod. Heal. Inform.* (2025) 203–210.
- [9] Mohammad Abobaker Khalifa, Mudassar Ali, Muhammad Naeem, Buoyant airborne turbines in b5g/6 g wireless networks: Opportunities, challenges, applications, and future directions, *Comput. Electr. Eng.* 111 (2023) 108962.
- [10] Rahul Ajmeria, Mayukh Mondal, Reya Banerjee, Tamesh Halder, Palla Kumar Deb, Debasish Mishra, Pravanjan Nayak, Sudip Misra, Surjya Kanta Pal, Debasish Chakravarty, A critical survey of eeg-based bci systems for applications in industrial internet of things, *IEEE Commun. Surv. Tutorials* 25 (1) (2022) 184–212.
- [11] Sri Harsha Kurra, Rama Krushna Rath, S.R. Sreeja, 2Dp-fhs: 2d pareto optimized fog head selection for multiple eeg healthcare data analysis and computations, in: *International Conference on Advances in Computing and Data Sciences*, Springer, 2024, pp. 58–68.
- [12] Arash Bozorgchenani, Daniele Tarchi, Giovanni Emanuele Corazza, Centralized and distributed architectures for energy and delay efficient fog network-based edge computing services, *IEEE Trans. Green Commun. Netw.* 3 (1) (2019) 250–263.
- [13] Racha Gouareb, Vasilis Friderikos, Abdol-Hamid Aghvami, Virtual network functions routing and placement for edge cloud latency minimization, *IEEE J. Sel. Areas Commun.* 36 (10) (2018) 2346–2357.
- [14] Ammar Awad Mutlag, Mohd Khanapi Abd Ghani, N al Arunkumar, Mazin Abed Mohammed, Othman Mohd, Enabling technologies for fog computing in healthcare iot systems, *Future Gener. Comput. Syst.* 90 (2019) 62–78.
- [15] Indranil Sarkar, Mainak Adhikari, Neeraj Kumar, Sanjay Kumar, Dynamic task placement for deadline-aware iot applications in federated fog networks, *IEEE Internet Things J.* 9 (2) (2022) 1469–1478.
- [16] Hafiz Farooq Ahmad, Wajid Rafique, Raihan Ur Rasool, Abdulaziz Alhumam, Zahid Anwar, Junaid Qadir, Leveraging 6G, extended reality, and iot big data analytics for healthcare: A review, *Comput. Sci. Rev.* 48 (2023) 100558.

- [17] Syed Rehan Abbas Jafri, Tehreem Hamid, Rabia Mahmood, Muhammad Asjad Alam, Talha Rafi, Muhammad Zeeshan Ul Haque, Muhammad Wasim Munir, Wireless brain computer interface for smart home and medical system, *Wirel. Pers. Commun.* 106 (4) (2019) 2163–2177.
- [18] Wai-Chi Fang, Kai-Yen Wang, Nicolas Fahier, Yun-Lung Ho, Yu-De Huang, Development and validation of an eeg-based real-time emotion recognition system using edge ai computing platform with convolutional neural network system-on-chip design, *IEEE J. Emerg. Sel. Top. Circuits Syst.* 9 (4) (2019) 645–657.
- [19] Aqeel Thamer Jawad, Rihab Maaloul, Lamia Chaari, A comprehensive survey on 6 g and beyond: Enabling technologies, opportunities of machine learning and challenges, *Comput. Netw.* 237 (2023) 110085.
- [20] Albena Mihovska, Atanas Vlahov, Vladimir Poulkov, 6G-based intelligent, context-aware, and trustworthy user-centric healthcare applications, in: 2024 Wireless Telecommunications Symposium, (WTS), 2024, pp. 1–6.
- [21] Amit Dolas, Roshan Bodile, Aryan Kaushik, Amandeep Kaur, Rohit Singh, Periklis Chatzimisios, Integrated ai and 6 g driven e-health: Enabling design, challenges, and future prospects, in: 2024 IEEE Conference on Standards for Communications and Networking, (CSCN), 2024, pp. 371–376.
- [22] Saiqin Long, Ying Zhang, Qingyong Deng, Tingrui Pei, Jinzhi Ouyang, Zhihua Xia, An efficient task offloading approach based on multi-objective evolutionary algorithm in cloud-edge collaborative environment, *IEEE Trans. Netw. Sci. Eng.* 10 (2) (2023) 645–657.
- [23] Fei Xu, Yue Xie, Yongyong Sun, Zengshi Qin, Gaojie Li, Zhuoya Zhang, Two-stage computing offloading algorithm in cloud-edge collaborative scenarios based on game theory, *Comput. Electr. Eng.* 97 (2022) 107624.
- [24] Amir Masoud Rahmani, Jawad Tanveer, Farhad Soleimanian Gharehchopogh, Shakiba Rajabi, Mehdi Hosseinzadeh, A novel offloading strategy for multi-user optimization in blockchain-enabled mobile edge computing networks for improved internet of things performance, *Comput. Electr. Eng.* 119 (2024) 109514.
- [25] Zhifu Gong, Jingyi Ji, Pengyuan Tong, Ahmed Sayed M. Metwally, Ashit Kumar Dutta, Joel J.P.C. Rodrigues, Ummul Hanan Mohamad, Smart urban planning: Intelligent cognitive analysis of healthcare data in cloud-based iot, *Comput. Electr. Eng.* 110 (2023) 108878.
- [26] Ashu Taneja, Adi Alhudhaif, Shtwai Alsubai, Abdullah Alqahtani, A novel multiple access scheme for 6 g assisted massive machine type communication, *IEEE Access* 10 (2022) 117638–117645.
- [27] Adyson Maia, Akram Boutouchent, Youcef Kardjadja, Manel Gherari, Ece Gelal Soyak, Muhammad Saqib, Kacem Boussekar, Idil Cilbir, Sama Habibi, Soukaina Ouledsidi Ali, et al., A survey on integrated computing, caching, and communication in the cloud-to-edge continuum, *Comput. Commun.* 219 (2024) 128–152.
- [28] Mekala Ratna Raju, Sai Krishna Mothku, Manoj Kumar Somesula, Drl-based task scheduling scheme in vehicular fog computing: Cooperative and mobility aware approach, *Ad Hoc Networks* 173 (2025) 103819.
- [29] Vahideh Hayyolalam, Öznur Özkasap, Cbwo: A novel multi-objective load balancing technique for cloud computing, *Future Gener. Comput. Syst.* 164 (2025) 107561.
- [30] P. Suresh, P. Keerthika, R. Manjula Devi, G.K. Kamalam, K. Logeswaran, Kishor Kumar Sadasivuni, K. Devendran, Optimized task scheduling approach with fault tolerant load balancing using multi-objective cat swarm optimization for multi-cloud environment, *Appl. Soft Comput.* 165 (2024) 112129.
- [31] Philipp Zagar, Vishnu Ravi, Lauren Aalami, Stephan Krusche, Oliver Aalami, Paul Schmiedmayer, Dynamic fog computing for enhanced llm execution in medical applications, *Smart Health* 36 (2025) 100577.
- [32] Ying-Yi Hong, Francisco I. Alano, Yih-der Lee, Chia-Yu Han, Edge-fog computing-based blockchain for networked microgrid frequency support, *IEEE Access* 13 (2025) 121720–121734.
- [33] Prasanthi Konduru, N.P. Nethravathi, Secure and energy-efficient routing protocol based on micro-segmentation and batch authentication, *Comput. Netw.* 248 (2024) 110293.
- [34] Manish Snehi, Abhinav Bhandari, Jyoti Verma Foggier skies, Clearer clouds: A real-time iot-ddos attack mitigation framework in fog-assisted software-defined cyber-physical systems, *Comput. Secur.* 139 (2024) 103702.
- [35] Abhishek Hazra, Mainak Adhikari, Tarachand Amgoth, Satish Narayana Srirama, Stackelberg game for service deployment of iot-enabled applications in 6G-aware fog networks, *IEEE Internet Things J.* 8 (7) (2021) 5185–5193.
- [36] Valentina Cacchiani, Manuel Iori, Alberto Locatelli, Silvano Martello, Knapsack problems - an overview of recent advances. part ii: Multiple, multidimensional, and quadratic knapsack problems, *Comput. Oper. Res.* 143 (2022) 105693.
- [37] Leonardo Lamanna, Renata Mansini, Roberto Zanotti, A two-phase kernel search variant for the multidimensional multiple-choice knapsack problem, *European J. Oper. Res.* 297 (1) (2022) 53–65.
- [38] Edara Sri Sai Kaushal, Rama Krushna Rath, S.R. Sreeja, Abhishek Hazra, 3D-pos: 3d pareto optimized head selection for fog-enabled smart eeg healthcare iot, in: 2025 IEEE 6th India Council International Subsections Conference, INDISCON, 2025, pp. 1–6.
- [39] Navid Khaledian, Shiva Razzaghzadeh, Zeynab Haghbayan, Marcus Völpl, Hybrid markov chain-based dynamic scheduling to improve load balancing performance in fog-cloud environment, *Sustain. Comput. Inform. Syst.* 45 (2025) 101077.
- [40] Niva Tripathy, Sampa Sahoo, Suwendu Chandan Nayak, Cheng-Chi Lee, An efficient task offloading approach to reduce energy consumption in fog architecture, *Comput. Electr. Eng.* 128 (2025) 110648.
- [41] Ali Kadhun Idrees, Sara Kadhun Idrees, Raphael Couturier, Tara Ali-Yahiya, An edge-fog computing-enabled lossless eeg data compression with epileptic seizure detection in iomt networks, *IEEE Internet Things J.* 9 (15) (2022) 13327–13337.
- [42] Sara Ghanavati, Jemal Abawajy, Davood Izadi, An energy aware task scheduling model using ant-mating optimization in fog computing environment, *IEEE Trans. Serv. Comput.* 15 (4) (2020) 2007–2017.
- [43] Mohammed Al-Khafajiy, Thar Baker, Hilal Al-Libawy, Zakaria Maamar, Moayad Aloqaily, Yaser Jararweh, Improving fog computing performance via fog-2-fog collaboration, *Future Gener. Comput. Syst.* 100 (2019) 266–280.
- [44] Indranil. Sarkar, Sanjay. Kumar, Deep learning-based energy-efficient computational offloading strategy in heterogeneous fog computing networks, *J. Supercomput.* 78 (13) (2022) 15089–15106.
- [45] Ismail Zahraddeen Yakubu, M. Murali, An efficient meta-heuristic resource allocation with load balancing in iot-fog-cloud computing environment, *J. Ambient. Intell. Humaniz. Comput.* 14 (3) (2023) 2981–2992.
- [46] Ralph G. Andrzejak, Klaus Lehnertz, Florian Mormann, Christoph Rieke, Peter David, Christian E. Elger, Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: Dependence on recording region and brain state, *Phys. Rev. E* 64 (6) (2001) 061907.